



RESEARCH ARTICLE

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Data-Driven Reduced Order Modeling for Warm Rain Microphysics

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Key Points:

- We introduce and compare three techniques for a data-driven latent space representation of droplet size distributions under coalescence
- Data from large eddy simulations with superdroplet microphysics are used to train and evaluate methods offline
- The simplest representation of time evolution that is based on polynomials generalizes best to out-of-sample data

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Accurately representing the formation of precipitation due to coalescence of cloud droplets remains a major challenge for atmospheric models. This study introduces and compares three data-driven approaches for modeling the time evolution of droplet size distributions within a reduced-order latent space: a polynomial-based Sparse Identification of Nonlinear Dynamics (SINDy) framework, numerical integration of a neural-network-predicted time derivative, and direct prediction of state at the next timestep via neural network. All three methods are coupled and simultaneously trained with an autoencoder that discovers the latent space variables. We train with high-fidelity data from large eddy simulations equipped with the superdroplet method, which captures the stochastic nature of the coalescence process. Our results show that while all three approaches can successfully emulate the nonlinear behavior of coalescence, the simplest, a polynomial-based SINDy representation, generalizes most robustly to out-of-sample cloud regimes, achieving strong performance and physical consistency. We further quantify model uncertainty using conformal prediction intervals, thereby providing rigorous estimates of predictive confidence across the modeling pipeline. These findings highlight that simple time evolution representations, when coupled with the correct prognostic variables, are a promising approach to advancing data-driven microphysics parameterizations toward operational atmospheric modeling.

Plain Language Summary Clouds produce rain through tiny water droplets colliding and merging, but simulating this process in weather and climate models is extremely challenging. We demonstrate that machine learning can create efficient and accurate simplified representations of these collisions when trained on detailed and realistic data. By comparing these techniques, we find that the simplest method based on polynomials does the best job of predicting how droplets evolve, even when tested on new types of clouds. Our approach produces accurate representations of rain formation, which can improve weather forecasts and climate predictions.

1. Introduction

Warm rain formation in clouds, driven by collisional coalescence of cloud droplets, remains notoriously difficult to represent accurately in atmospheric models. The formation of precipitation through droplet coalescence is especially challenging to parameterize in kilometer-scale and larger domains due to complex nonlinear interactions between droplets at the microscale, in addition to interactions with other unresolved processes such as turbulent entrainment and detrainment. Traditional approaches to representing coalescence processes rely on artificial categorization of hydrometeors into bulk classes of cloud droplets and rain (Seifert & Beheng, 2001, e.g.), introducing both (a) structural error, in the choice of prognostic variable; and (b) parametric uncertainty in the necessitated conversion rates between these categories (Grabowski et al., 2019; Morrison, van Lier-Walqui et al., 2020).

Recent growth in computational resources has enabled the use of more sophisticated approaches to modeling cloud microphysics at cloud-resolving scales. Bin microphysics, for instance, explicitly resolves the evolution of a continuous droplet size distribution (DSD), but remains computationally prohibitive for operational weather and climate models, suffers from numerical broadening during condensation and advection, and cannot capture certain stochastic behaviors inherent to droplet populations (Grabowski et al., 2019; Khain et al., 2015; Lebo & Morrison, 2013). The superdroplet method (SDM) offers an alternative probabilistic approach that simulates the stochastic nature of droplet interactions in a physically consistent Lagrangian framework (Andrejczuk et al., 2008; Li et al., 2022; S. Shima et al., 2009; Unterstrasser et al., 2020). While the high-fidelity SDM provides

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valuable insights into cloud microphysics at the process level, it also remains too computationally demanding for direct implementation in regional or global atmospheric models.

Data-driven approaches to represent subgrid-scale processes offer a promising pathway to bridge the gap between computational feasibility and physical accuracy. Machine learning emulators have been increasingly applied to represent complex unresolved processes such as thermodynamics (Brenowitz & Bretherton, 2018; Rasp et al., 2018), convection (Brenowitz et al., 2020), and entrainment (Christopoulos et al., 2024), for example. However, black-box neural network emulators are typically trained offline and often lack interpretability, physical consistency, and robustness when deployed online (Beucler et al., 2021; Seifert & Rasp, 2020). Data-driven approaches to microphysical parameterizations have mainly relied on emulating processes of auto-conversion, accretion, and self-collection within traditional categorical structures (Gettelman et al., 2021; Seifert & Rasp, 2020; Sharma & Greenberg, 2025), preserving interpretability but perpetuating the structural deficiencies of this prognostic variable set. These limitations highlight a need for data-driven approaches that capture essential microphysical processes in a generalizable and interpretable framework that goes beyond bulk microphysics.

Building on the foundational work on Lamb et al. (2024) and Champion et al. (2019), we propose a novel reduced-order modeling framework for warm rain microphysics with an emphasis on jointly learning prognostic latent variables alongside nonlinear governing time evolution equations. Lamb et al. (2024) demonstrated that a nonlinear mapping between a binned DSD and as few as three latent variables is effective to compress and recover the important complexity of the coalescence of DSDs ranging from cloud, to bimodal, to rain-dominant droplet populations. Related efforts to discover optimal prognostic variable sets through Non-negative Weighted Integrals (Huang et al., 2025) or non-categorical moment representations (Igel et al., 2022; Morrison et al., 2019, 2025) also support this finding, suggesting that future parameterizations need not follow a cloud-rain paradigm. In a different context of dynamical systems and fluid mechanics, Champion et al. (2019) established a framework for discovering parsimonious nonlinear dynamical models in a data-driven latent space using sparse identification of nonlinear dynamics (SINDy) (Brunton et al., 2016).

Our approach combines and extends these data-driven latent space methodologies to address the challenge of representing droplet coalescence within this reduced-order modeling context. Specifically, we introduce and compare three data-driven approaches to model the time evolution of a nonlinear latent space, providing insights into time evolution representations that complement and extend existing work on microphysical state representations. While many other efforts to learn microphysical equations rely on bin microphysics for training data (Gettelman et al., 2021; Lamb et al., 2024; Morrison, van Lier-Walqui et al., 2020), our approach utilizes SDM simulations to inform not only process rates—as in Seifert and Rasp (2020) and Sharma and Greenberg (2025)—but also the prognostic variables themselves. This pipeline ensures that the learned reduced-order model (ROM) is not only trained on high-fidelity data but also is robust to stochastic variability in DSDs that arise from collisional coalescence. Our focus on the process rates of cloud microphysics, with coalescence as the first step, represents a step toward practical implementation of a fully data-driven ROM for microphysics and offers key insights into machine-learning emulator structures that may inform data-driven parameterization of other subgrid-scale processes.

The core objectives of this study are threefold:

1. We seek to reduce both structural and parametric uncertainty in warm-rain microphysics without increasing computational complexity—specifically, the memory and computational burden of additional prognostic variables—beyond bulk approaches.
2. We explore the balance between accuracy, interpretability, and generalizability across different time evolution operators, prioritizing the discovery of latent-space governing equations that capture essential characteristics of droplet coalescence while maintaining robustness across a range of cloud conditions.
3. We aim to provide guidance on the optimal level of structural complexity in the reduced-order modeling of cloud microphysics, which will be extended to represent additional microphysical processes in future work.

The remainder of this paper is organized as follows. Section 2 introduces our reduced-order modeling framework, including the three data-driven time evolution operator structures and training strategy. Section 3 describes the superdroplet simulations used for training, validation, and testing, as well as specifics of the neural network structures and uncertainty quantification methods. Section 4 presents the results of our reduced-order models in an offline setting, and Section 5 further evaluates the generalizability of the models to out-of-sample cloud

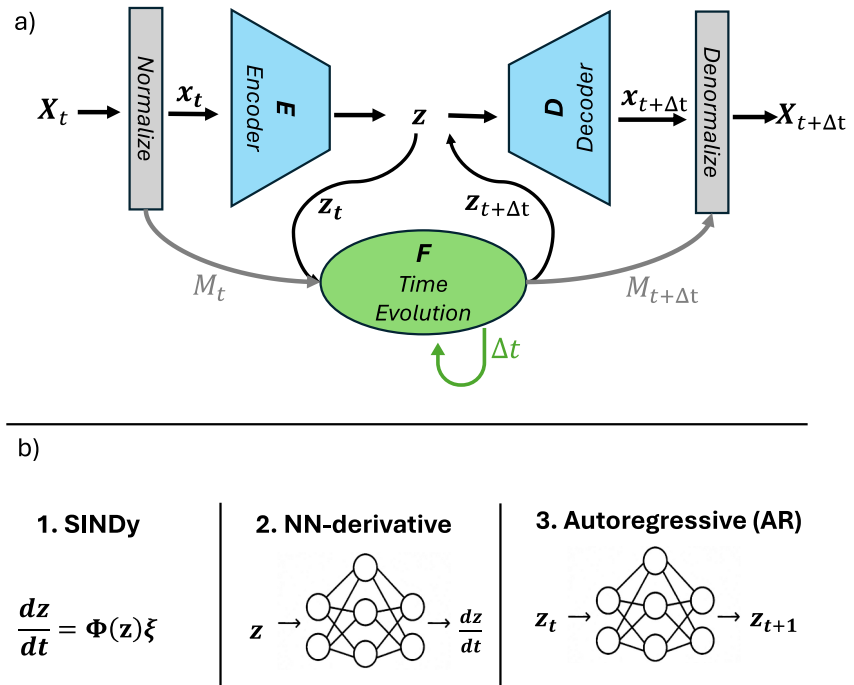


Figure 1. (a) The three main components of the reduced-order model architecture: the normalizer/denormalizer, the encoder/decoder, and the time evolution; and (b) three alternative architectures for the time evolution model F considered in this study. The subscripts t and $t + \Delta t$ indicate the initial and forward states, respectively, for the unnormalized droplet size distribution (DSD) ($\mathbf{X}$), normalized DSD ($\mathbf{x}$), DSD encoding ($\mathbf{z}$), and total mass (M). For the Sparse Identification of Nonlinear Dynamics model, $\Theta(\mathbf{z})$ are the library terms (order-2 polynomials, in our case) and ξ are the coefficients.

conditions and presents empirical uncertainty intervals. Finally, Section 6 discusses the implications of our findings for future parameterization development and concludes with a set of recommendations and next steps toward integrating reduced-order models of subgrid-scale microphysics in operational atmospheric models.

2. Reduced-Order Modeling Framework

Our reduced-order modeling (ROM) framework combines a nonlinear autoencoder (AE) with a suite of data-driven time evolution models. Unlike more traditional Physics-Informed Neural Networks, which penalize non-conservation in the loss function, we utilize a mass-rescaling technique designed to enable strict enforcement of conservation laws as desired. Specifically, we use the convention that $\mathbf{X}(t)$ is a log-space particle mass distribution $dm/d \ln(r)$ within an atmospheric volume with units of [mass of liquid/air volume/bin width $d \ln(r)$], where r refers to particle radius. M is the total mass of this DSD, and $\mathbf{x}$ is the DSD $\mathbf{X}$ normalized by this total mass:

$$M(t) = \sum_{i=0}^{N_{bin}} X_i(t) d \ln(r)_i, \quad (1)$$

$$x_i(t) = \frac{X_i(t)}{M(t)} \text{ s.t. } \sum_{i=0}^{N_{bin}} x_i(t) = 1. \quad (2)$$

In addition to this normalization step, the overall ROM pipeline consists of three key components (pictured in Figure 1):

1. Encoder $E : \mathbf{x} \rightarrow \mathbf{z}$. Maps the N_{bin} -dimensional discrete binned DSD $\mathbf{x}$ normalized by total liquid mass M into a small $(N_{latent} - 1)$ -dimensional DSD-encoding $\mathbf{z}$. The full latent state $\mathbf{Z}$ includes M : $\mathbf{Z} = \mathbf{z} ++ M$, where $++$ indicates concatenation of the total liquid mass M to the encoded DSD vector $\mathbf{z}$.
2. Decoder $D : \mathbf{z} \rightarrow \mathbf{x}$. Maps the DSD encoding back to a normalized, binned DSD. The DSD can be denormalized through multiplication with M from the latent vector.

3. Evolution operator $F : \mathcal{Z}(t) \rightarrow \mathcal{Z}(t + \Delta t)$. Evolves the full latent state forward in time using one of three approaches, summarized in Figure 1b. F can represent either a time-rate-of-change $\frac{d\mathcal{Z}}{dt}$, or an autoregressive operator over a fixed time step as $\mathcal{Z}_{t+1} = F(\mathcal{Z}_t)$.
4. Normalization: $\mathbf{X} \rightarrow [\mathbf{x}, M]$; and denormalization: $[\mathbf{x}, M] \rightarrow \mathbf{X}$. Normalizes the dimensioned binned particle mass distribution $\mathbf{X}(t)$ by the total mass of that distribution, $M(t)$, producing a dimensionless DSD state vector $\mathbf{x}(t)$ whose components sum to unity.

The first $(N_{\text{latent}} - 1)$ components of the latent space $\mathcal{Z}$ describes the shape of the DSD, and the final component M represents its magnitude. The normalized DSD can be encoded to its latent space and approximately reconstructed via the composition of the encoder and decoder so that

$$\mathbf{x} \approx D(E(\mathbf{x})). \quad (3)$$

The operator F evolves the full latent state in time and is specified such that any number of timesteps Δt may be taken within the latent space without requiring decoding or re-encoding between timesteps.

While coalescence conserves the total liquid mass M , other microphysical processes such as condensation and evaporation are expected to impact not only the shape of a DSD, but also its total mass as water changes phase between liquid and vapor. In the case of coalescence, M is included both as an input that impacts the rate of change of the DSD encoding $\mathbf{z}$ and as a rate-of-change output from the evolution operator; however, strict mass conservation can be enforced in this case by discarding the predicted rate-of-change in M . In the case of other non-liquid-conserving subgrid-scale processes such as condensation/evaporation, total water mass conservation can be enforced within a computational domain element by specifying

$$M(t + \Delta t) + \rho_a q_v(t + \Delta t) = M(t) + \rho_a q_v(t), \quad (4)$$

where ρ_a is the gridcell air density and q_v is the gridcell water vapor concentration (units of kg water vapor/kg air).

The present study focuses on the process of collisional coalescence at the subgrid scale, leaving other microphysical processes—such as condensational growth, aerosol nucleation, or ice processes, as well as advection of quantities between gridboxes and coupling to other subgrid-scale (SGS) parameterizations—to future work. The collisional coalescence process conserves total liquid mass, providing a simple opportunity to evaluate the performance of the evolution operator F in learning not only changes to the normalized DSD latent space, but also a conserved physical quantity. In the case of droplet coalescence and other mass-conserving processes, conservation can be directly enforced during the denormalization step and/or by overwriting the total mass element of the latent state, regardless of the time evolution of model predictions. However, for other processes involving phase change or transport, total liquid water is not conserved, necessitating stronger reliance on either predictions by the data-driven model, or alternative physics-based constraints to correct these normalizing quantities.

2.1. Autoencoder: A Feed-Forward Neural Network

The encoder and decoder structures used by our AE are inspired by the convolutional autoencoder developed by Lamb et al. (2024), but with a simpler structure. While Lamb et al. (2024) encode and decode two DSD channels (mass and number density) with a series of three convolutional layers, our approach applies only to the mass-resolved DSD using four fully connected feed-forward layers with halving/doubling of the vector dimension within each layer of the encoder/decoder (respectively). The decision to use fully connected rather than convolutional layers follows from our finding that training a feed-forward network is more efficient and more stable, whereas attempts to use a convolutional neural network frequently led to plateaus in the loss function corresponding to a constant-valued encoding and reconstruction of DSDs. In this study, we use an input-vector size of $N_{\text{bin}} = 64$ logarithmically spaced bins over a size range of 1 μm –5 mm radius with an encoding dimension size of 3, following from the finding in Lamb et al. (2024) that the intrinsic dimension of coalescence is 3; this finding was additionally confirmed in optimization studies which varied the latent dimensionality. Although the SDM is used to generate training data, we choose to design our ROM to operate in the space of a binned size distribution derived from the SDM results in order to ensure that target data state has a constant size vector representation (64 bins, rather than a variable number of superdroplets) across samples and across time.

2.2. Evolution Operator Structures

We implement and compare three distinct approaches for modeling the time evolution of the latent variables. The first two approaches learn a data-driven representation of the time-derivative of the latent space, $\frac{d\mathcal{Z}}{dt}$, which is then evolved forward in time using the Runge-Kutta 45 algorithm in SciPy (Virtanen et al., 2020) with stabilizing limiters (see 2.2.4). The third autoregressive method instead learns the discrete-time evolution of the latent space with a fixed time step.

2.2.1. SINDy (Sparse Identification of Nonlinear Dynamics)

The SINDy approach expresses latent time evolution as a system of coupled ordinary differential equations (ODEs) (Brunton et al., 2016; Champion et al., 2019):

$$\frac{d\mathcal{Z}}{dt} = \Theta(\mathcal{Z})\xi, \quad (5)$$

where $\Theta(\mathcal{Z})$ is a vector of user-defined candidate functions and ξ is the matrix of ML-learned coefficients; multiplication of $\Theta(\mathcal{Z})$ and ξ leads to a system of coupled ODEs. Specifically, this work utilizes polynomials up to order 2, including a constant, 4 first-order linear terms, and 10 second-order quadratics including cross-terms, for a total of 60 unknown coefficients (15 for each of the 4 latent variables).

While the original SINDy algorithms learn a sparse coefficient matrix via least squares, our approach reframes this paradigm as a bias-free single linear layer so that the layer's weights correspond to the ODE coefficients ξ . A core component of the original SINDy technique is sparsity of the ODE system, or a large number of zero-values in ξ promoted through a regularizing penalty term on the coefficient matrix as part of the overall least-squares loss function (Brunton et al., 2016). Later iterations on SINDy use a sequential-thresholding technique that periodically removes the smallest non-zero coefficient values (Champion et al., 2019). The use cases of SINDy presented in this work use comparatively small function libraries based on at most 2nd-order polynomials, thus sparsity is not a core constraint in our application, and neither regularization nor sequential thresholding were used.

2.2.2. Neural-Network-Driven Time Derivative

Our second approach utilizes a more traditional machine learning scaffolding, where a multilayer neural network NN is substituted to learn the time derivatives of the prognostic variables $\mathcal{Z}$:

$$\frac{d\mathcal{Z}}{dt} = NN(\mathcal{Z}). \quad (6)$$

This approach is comparable to direct learning of warm-rain processes such as autoconversion or accretion (Gettelman et al., 2021; Sharma & Greenberg, 2024), except that our version simultaneously learns optimal prognostic DSD encoding $\mathbf{z}$ rather than assuming two-category cloud or rain moments a priori. The NN-derivative model in this work uses three-fully connected layers with ReLU activation functions; the size of each layer determined through hyperparameter optimization (Section 3.4).

2.2.3. Autoregression

Our final approach directly predicts the next discrete time state with a single fixed time step using an autoregressive network (AR), rather than learning a continuous time derivative:

$$\mathcal{Z}_{t+1} = AR(\mathcal{Z}_t). \quad (7)$$

Previous studies suggest that autoregression may result in more stable and robust predictors compared with derivative-learning approaches (Lamb et al., 2025; McCabe & Freedman, 2023). That being said, reliance on a constant time step may be limiting when applying such a parameterization to multiple physical scales, such as climate modeling versus a single-cloud study. While more complex structures such as recurrent neural networks and transformers may utilize multiple prior state vectors as inputs to predict the next, we investigate only a single-

lagged approach, where both training and prediction occur across a single time step at a time. Our AR structure uses a neural network with 3 hidden layers and ReLU activation; the size of each layer is again determined through hyperparameter optimization (Section 3.4).

2.2.4. Stability and Limiters

While the AR model is freely evolving, the NN-derivative and SINDy are not necessarily stable ODE systems, and can lead to exponential growth of the latent variables from certain initial conditions. In order to prevent such behavior, we adopt the common technique in numerical weather prediction of enforcing limiters. For every latent variable $\mathcal{Z}_i$ in vector $\mathcal{Z}$, a maximum allowable interval $[\mathcal{Z}_{i,\min}, \mathcal{Z}_{i,\max}]$ is determined based on the range of encoded training data. Within the ODE solution, the derivative for a particular latent component $\mathcal{Z}_i$ is set to zero if its instantaneous value falls outside of this interval. Alternatively, a high-order dissipative or damping term with a small prefactor may be added to the ODE system in order to drive large-magnitude solutions toward a stable region (“trapping”) (Kaptanoglu et al., 2021). For instance, when using quadratic SINDy polynomials as in our applications, cubic damping prevents runaway solutions. This practice is less common than zero-th or first-order limiters in subgrid scale parameterizations, where model performance is dependent on all components remaining well-behaved, and thus a larger degree of SGS error may be tolerated.

2.3. Optimization Strategy & Loss Function

A core feature of our ROM framework is the simultaneous training of the autoencoder components and the time evolution model through a joint loss function. This approach ensures that the encoded latent space not only preserves information key to reconstructing DSD, but that it also identifies a low-dimensional manifold on which DSD evolution under coalescence can be optimally represented. The loss function L is a weighted sum of three components:

$$L = L_{recon} + w_{dx}L_{dx} + w_{dz}L_{dz}, \quad (8)$$

which address the following questions:

1. Reconstruction L_{recon} : How well can the encoder-decoder compress a DSD to a latent vector, and then reconstruct the same DSD?
2. DSD-evolution L_{dx} : How well does the entire pipeline reproduce observed rates of change in $\mathbf{x}$?
3. Latent-evolution L_{dz} : Within the latent space $\mathcal{Z}$, how well does F reproduce observed rates of change?

Including separate reconstruction and DSD evolution loss terms incentivizes separate learning of the DSD structure by E and D , and learning of the time evolution by F . Only L_{dz} penalizes the model for violating mass conservation principles, as M is an element of the latent space but not captured in the normalized DSDs. Although we can in principle guarantee mass conservation in the context of a dynamic such as coalescence, the ability of the time evolution operator to correctly learn rates of change of the physical variables will be important in the context of phase change or advection in future work. This latent-evolution loss similarly rewards F for learning a precise and stable time evolution model within the latent space where the time-stepping actually occurs. The weights w_{dx} and w_{dz} prioritize each loss function component relative to reconstruction and are tuned hyperparameters, as described in Section 3.4.

For all models, the reconstruction loss evaluated on a normalized DSD $\mathbf{x}$ is defined as the Kullback–Leibler (KL) divergence (Kullback & Leibler, 1951) between each sample and its reconstruction:

$$L_{recon}[E, D](\mathbf{x}) = KL(\mathbf{x}, D(E(\mathbf{x}))). \quad (9)$$

When F takes the form of SINDy or a NN-derivative, the time evolution losses are computed as a mean-squared-error (MSE) (represented as the square of the 2-norm $\|\cdot\|_2^2$) between the predicted and observed rates of change $d\mathbf{x}/dt$, projected into normalized DSD space or into latent space as in Champion et al. (2019):

$$L_{dx}[E, D, F]\left(\mathbf{x}, \frac{d\mathbf{x}}{dt}\right) = \left\| \frac{d\mathbf{x}}{dt} - \nabla_{\mathbf{z}} D(\mathbf{z}) \cdot F(\mathcal{Z})_{1:N_{latent}-1} \right\|_2^2, \quad (10)$$

$$L_{dz}[E, F]\left(\mathbf{x}, \frac{d\mathbf{x}}{dt}\right) = \left\| \left(\nabla_{\mathbf{x}} E(\mathbf{x}) + \mathbf{0} \right) \cdot \left(\frac{d\mathbf{x}}{dt} + \mathbf{0} \right) - F(\mathbf{z}) \right\|_2^2, \quad (11)$$

with $\mathbf{z} = E(\mathbf{X})$ throughout. In Equation 10, the dot-product $(\cdot)$ between the gradient of the decoder and the predicted time derivative represents projection of a natively latent-space component $F(\mathbf{Z})$ into DSD space. Likewise, in Equation 11, the dot product between the encoder gradient and the observed time-rate-of-change projects the latter data into the latent space with zero rate-of-change in the final component, M .

When F is autoregressive, the time evolution losses are computed between consecutive discrete time predictions t and $t + 1$. Hence, the DSD-space loss term utilizes KL-divergence rather than MSE-loss:

$$L_{dx}[E, D, F](\mathbf{x}_t, \mathbf{x}_{t+1}) = KL(\mathbf{x}_{t+1}, D(F(E(\mathbf{x}_t)))), \quad (12)$$

$$L_{dz}[F](\mathbf{z}_t, \mathbf{z}_{t+1}) = \|\mathbf{z}_{t+1} - F(\mathbf{z}_t)\|_2^2. \quad (13)$$

While these loss terms could also be applied in the context of the SINDy or NN-derivative, they would be dependent in such cases on the choice of time-step and ODE solver strategy. Thus we prefer to operate in the space of time derivatives in these native time-derivative evolution operators.

In addition to the innate training loss metrics (KL Divergence, MSE), we characterize the difference between predicted and actual DSDs according to the Wasserstein metric (Villani, 2003), often referred to as the Earth Mover's distance. This metric is a symmetric measure of the difference between two distributions and thus weights all portions of the DSD equally, whereas KL divergence used in training is an asymmetric measure based on relative entropy. KL divergence is widely used and innately supported by deep learning packages while Wasserstein distance is not, thus Wasserstein is reserved for evaluation and for hyperparameter optimization.

3. Data and Methods

3.1. Large Eddy Simulation With Superdroplet Method

Data for this study are produced using the Energy Research and Forecasting model (ERF) equipped with Lagrangian superdroplet microphysics. ERF is a large eddy/regional scale atmospheric simulation code designed for the latest high performance computing architectures (Beydoun et al., 2026; Lattanzi et al., 2025). It solves the fully compressible three-dimensional moist Euler/Navier-Stokes equations (with gravitational forcing/buoyancy) on a staggered Arakawa C-grid. The superdroplet method (SDM) evolves the droplet mass and size via explicit handling of condensation/evaporation and applies the Monte-Carlo coalescence method developed by S. Shima et al. (2009) based on the ambient flow conditions.

Training, validation, and testing data are processed from two idealized simulation setups, both of which are briefly summarized here. The first setup is the isolated congestus cloud: initially developed by Lasher-Trapp et al. (2005), the congestus setup applies spatially Gaussian sensible and latent heat fluxes to a spun-up cloudy boundary layer, producing a cloud that deepens, precipitates, and dissipates. We sample three different time states of the congestus simulation (Figure 2): a condensation-dominant state at 4800s, a coalescence-dominant state at 5400s, and a sedimentation-dominant state at 7200s simulation time. The second case is the Rain in Cumulus over the Ocean (RICO) LES setup developed by van Zanten et al. (2011), sampled near its steady state after an initial 20 hr of spin-up. In both congestus and RICO, superdroplets are initialized as a bimodal distribution of ammonium bisulfate with the recommended modes and standard deviations taken from van Zanten et al. (2011). The congestus cloud case uses a fixed time step of 0.5 s and a grid of $128 \times 128 \times 200$ elements spaced at 50 m in all 3 dimensions, with 128 superdroplets per grid cell initially (64 per aerosol mode). The RICO simulation uses 100 m resolution in the horizontal and 40 m in the vertical with a $128 \times 128 \times 100$ grid cell domain, a fixed 0.3 s time step, and 128 superdroplets per grid cell initially (64 per aerosol mode).

In order to isolate the impacts of coalescence on the DSD, each data set begins from a restart state with all non-coalescence particle processes—advection, sedimentation, and phase-change—disabled. During this coalescence-only mode, the thermodynamic and velocity fields determined by ERF's dynamical core have no impact on the superdroplets, making this step equivalent to evolving coalescence within each gridbox

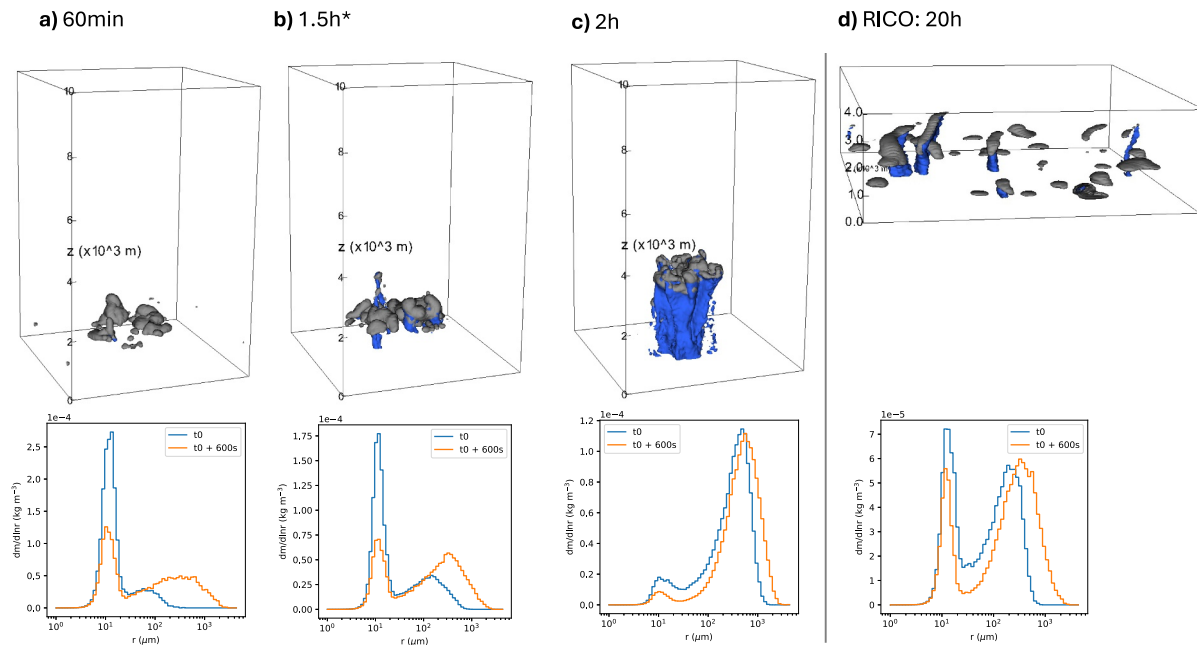


Figure 2. 3D contour snapshots (top row) for the beginning state t_0 of each of the four LES-SDM data sets used in this study. Contours correspond to 10^{-5} kg/kg cloud water mixing ratio (gray) and 10^{-4} kg/kg rain water mixing ratio (blue) using a cutoff of $40 \mu\text{m}$ radius to categorize superdroplets. Each case contains an illustration of the mean droplet size distribution across the domain (excluding gridboxes with insufficient liquid water; bottom row) at the start of each state (blue) and following 600s of coalescence-only evolution (orange). Cases (a–c) come from the congestus simulation, while case (d) uses the RICO simulation setup. * denotes the data set used for training and validation.

independently in an SDM box configuration. (As in S.-I. Shima et al. (2020), ERF-SDM only considers collisions between superdroplets within the same gridbox.) We run each modified simulation forward for 600 s with the native model time step, storing the unstructured superdroplet state every 10 s (approximately 30 GB per time step). Data are then coarse-grained to cubic gridboxes of size $200 \text{ m} \times 200 \text{ m} \times 200 \text{ m}$ ($200 \text{ m} \times 200 \text{ m} \times 40 \text{ m}$)—64x (4x) larger than the simulation grid spacing with an average of 8,192 (512) superdroplets per processed volume in the case of congestus (RICO)—over the full 600s. The clouds produced by RICO are typically shallower than the congestus cloud, hence the decision to utilize a smaller vertical grid for averaging. We store the DSD as the liquid mass represented by the superdroplets sorted into 64 logarithmically spaced radius bins ranging from $1 \mu\text{m}$ to 5 mm . The total liquid water mass of droplets greater than $1 \mu\text{m}$ radius is thus exactly conserved in this bin representation x , whereas the total number concentration or higher-order moments cannot be exactly reconstructed from x .

Training and validation data are taken from the congestus cloud state at 5400s, which experiences more diverse precipitation states and the largest mean changes to the DSD due to droplet coalescence (Figure 2). Random samples from the remaining data sets are then used for quantitative testing to assess the generalizability and uncertainty of the trained models. Only processed volumes with sufficient liquid water mixing ratio greater than 10^{-5} kg/kg are used to train and evaluate the coalescence ROM. For the congestus case beginning at 5400s, this results in a total of 618 DSDs that evolve independently under coalescence, from which we randomly sample 80% of the DSDs for training, reserving the remaining 20% for validation. The 4800 and 7200s congestus cases contain 1,300 and 12,238 total DSDs with sufficient liquid water content, respectively, and we sample 20% (260) and 2% (255) from these cases (respectively) at random for testing. The RICO case contains 3228 suitable DSDs from which we sample 10% (323) for testing. The RICO case is later separated into a 40%–60% calibration-testing split for uncertainty quantification. (See data summary in Table 2.) Only the 5400s congestus data set is seen by the ROMs during model training.

3.2. Two-Moment Bulk Scheme Comparison

In evaluating DSD reconstruction and end-to-end model performance, we include an example of a bulk moment-based microphysics scheme from Seifert and Beheng (2001) (SB2). This 2-moment scheme assumes a gamma

size distribution in mass for cloud droplets (radius $\leq 40\mu\text{m}$) and an exponential size distribution in diameter for rain drops. We select a fixed gamma shape parameter of $\nu = 1.0$ for the cloud DSD, and utilize the self-similarity solution derived in Seifert and Beheng (2001) for the Long (1974) coalescence kernel to solve for the time evolution of cloud and rain number and mass concentration, respectively. Number and mass concentrations are limited to positive values. As noted in Seifert and Beheng (2001), self-collection has a tendency to drive raindrop number concentration toward zero. This behavior was not found to be mitigated with the use of more sophisticated time steppers, thus we utilize an explicit Euler time step of 10 s to evolve the moments forward in time. The PSDs implied by the SB2 closure are plotted and evaluated at the cubic mean center of each radius bin.

3.3. Network Structure & Training

Specification, training, and exporting of machine learning structures in this study utilize the PyTorch library (Paszke et al., 2019). The encoder/decoder networks use feed-forward neural networks with 4 layers each. The first three layers of the encoder sequentially reduce the dimensionality by a factor of 2 (from 64, to 32, to 16 in our examples) with a fully connected layer and ReLU activation (Nair & Hinton, 2010). The final fully connected layer then maps to $N_{\text{latent}} - 1$ latent variables (recall that M is the final latent variable) with no nonlinearity. The decoder reverses this procedure with fully connected layers—after the final transform back to N_{bin} dimensions, the decoder applies a final softmax activation layer to ensure that the prediction $\mathbf{x}$ is a normalized DSD. Weights of the encoder and decoder are initialized from a Xavier-normal distribution (Glorot & Bengio, 2010) with zero bias. While Lamb et al. (2024) used a convolutional neural network to encode and decode their size distributions, we found that using a feed-forward network instead achieved comparable performance with more stable training and fewer plateaus.

Time evolution operators are tasked with learning time derivatives (SINDy, NN-derivative) or predicting the next state across a fixed time step (AR). Specifically, we sample DSDs every 50 s from the ERF-SDM data sets, computing the rate of change of normalized DSDs using a finite difference approximation for the derivative-learning operators, and simply using the 50-second states for the AR operator. Thus for each postprocessed simulation volume, five different times are seen during training. The learning time step of 50 s is selected to be half of the 100 s time step used in a large-scale cloud-resolving model “SCREAM” (Donahue et al., 2024). SINDy coefficients ξ are initialized to zero. The NN-derivative time evolution model uses 4 fully connected layers with SiLU activation (Ramachandran et al., 2017) on the hidden states using Kaiming-normal initialization (He et al., 2015) on all but the final layer, Gaussian initialization on the final layer, and zero bias initialization. The AR time evolution model follows the same structure but with an ELU activation function (Clevert et al., 2016) on hidden states and a sigmoid activation on the final layer, with Xavier-uniform initialization (Glorot & Bengio, 2010) and zero bias. The dimension of the hidden states in both of the latter methods are determined through hyperparameter optimization (Section 3.3).

While DSDs are normalized ($\mathbf{x} \rightarrow \mathbf{X}, M$) prior to encoding, we further rescale total liquid masses by the maximum value of liquid mass within the training set. All three AE-time evolution methods utilize early stopping with a patience of 50 epochs along with a scheduler that reduces the learning rate by a factor of 10 when a plateau is detected in the validation loss; model training is halted after approximately 100 epochs in all three cases (see Figures S11-3 in Supporting Information S1). Initial learning rates and training batch size are determined through hyperparameter optimization (described below). Training takes several minutes on an Apple M3 Pro chip.

3.4. Hyperparameter Optimization

We optimized the hyperparameters of each network separately using the open-source hyperparameter optimization Python library Optuna (Akiba et al., 2019). Optuna uses Bayesian optimization algorithms to search a network's hyperparameter space for the optimal values. While Optuna can handle numerical and categorical variables and is capable of optimizing many aspects of a network's architecture, we restricted our search to 3–7 key hyperparameters per network. This choice ensures that the search space is not too large and builds on adequate results found during preceding hand-tuning.

The hyperparameters we searched for are listed in Table 1 by network architecture. A hyperparameter was only excluded from the search space if it did not apply to a given network's architecture. As described in (Champion et al., 2019), w_{dx} is set to “slightly less than” $\sum_{i=1}^m \|\mathbf{x}_i\|_2^2 / \sum_{i=1}^m \|\hat{\mathbf{x}}_i\|_2^2$, where “slightly less than” is a scaling factor we call the “ w_{dx} metaweight.” w_{dx} is derived from w_{dx} as a factor of 100 smaller. The w_{dx} metaweight is therefore

the only tuning parameter that is optimized, which is why there is an asterisk (*) next to the checkmarks for w_{dx} and w_{dz} in Table 1.

The objective function that Optuna tried to minimize was the mean Wasserstein distance (Villani, 2003) between the true and predicted training set. This objective function is deliberately different from the training loss function (KL divergence and MSE losses), and optimizes for performance over the entire training set. For each network, we ran Optuna for 1000 iterations using the Tree-structured Parzen Estimator optimization algorithm (Bergstra et al., 2011) and the Hyperband pruning algorithm (Li et al., 2018). In each iteration, the model is trained for a maximum of only 30 epochs; a majority of the learning happens early in the training process, and as such, the relative performance of a given hyperparameter configuration is distinguishable from another. This pattern is noticeable enough that Optuna's trial pruner is often activated even though the total number of epochs in each trial is low. We used the resulting hyperparameters for each network to report the final results. Details of the exact values searched for and found by Optuna can be seen in the Supporting Information S1, though in general, learning rate and batch size were the most important hyperparameters for any study.

3.5. Prediction Intervals and Uncertainty

We compare uncertainty across various model components by computing conformal prediction (CP) intervals (Angelopoulos & Bates, 2022) following the framework introduced in Katona et al. (2025). Conformal prediction quantifies model output uncertainty by estimating the output error bound for a given confidence level $1 - \alpha$. By evaluating the data-prediction mismatch, called nonconformity scores, on held-out calibration data, conformal prediction provides an empirical guarantee that the distance between the prediction and value of a new unseen data point will with probability $1 - \alpha$ fall within the interval. Following Katona et al. (2025), we compute two-sided signed prediction intervals on normalized DSD predictions, bin-by-bin, and on latent variable predictions. In the case of DSDs, intervals can be computed bin-by-bin using the difference between the predicted and actual X_i for each DSD bin i . An example illustrating how these bin-by-bin intervals translate to the uncertainty on a DSD prediction is included in Figure S4 of Supporting Information S1. This framework allows us to separately compute intervals of the autoencoders and time evolution models, as well as of the end-to-end ROM output, in order to diagnose and compare uncertainty stemming from each component of the ROM pipeline.

Our particular application uses the “split conformal” variation of CP (Lei et al., 2017) in which data that were not used to train the model are split into “calibration” and “testing” data. The residuals between actual and predicted data in bin or latent space are computed on the calibration subset and then applied to the testing data. Because our 5400s congestus training-validation split has a limited supply of non-training data points, we instead apply CP to the RICO data set, utilizing 40% of the total data set for calibration and the remainder for testing. In Figure 7, we visualize the integrated predictive uncertainty as a function of time across different components of the model architecture. For DSD-valued predictions, the area between overprediction and underprediction bands is integrated over bins using the composite trapezoidal rule. For the latent variables, uncertainty is quantified using the Mahalanobis distance (Mahalanobis, 1936) to appropriately weight the differing range of various latent variables.

4. Model Performance and Characteristics

4.1. Latent Space & DSD Reconstruction

Figures 3 and 4 illustrate the reconstruction performance and latent space representation of the encoded true DSD data, without attempting to predict time evolution. Figure 3 demonstrates that the trained autoencoders achieve strong reconstruction skill, even with a relatively simple feed-forward architecture and single input channel (binned mass distribution). The autoencoder successfully encodes and reconstructs key DSD characteristics, including bimodal behavior and the mean droplet size and breadth of each mode. Reconstruction is worst for narrow or highly peaked distributions, particularly when small secondary modes are present. We illustrate such characteristic cases based on percentile performance: the n -th percentile case performs better than n percent of all DSDs in the validation set in terms of mean Wasserstein distance between the modeled and actual normalized DSD, averaged across the 600s trajectory. For example, in the 10th-percentile AE-SINDy case (meaning that 90% of cases have a lower Wasserstein distance), the autoencoder fails to reproduce the initial weak second mode, leading to large transient errors early in the trajectory when this weak rain mode likely accelerates initial coalescence (Figure 3c). All three time evolution formulations show similar patterns of reconstruction error across the validation set, consistent with the similar latent space structure observed in Figure 4. Indeed, commonalities

between the reconstruction performance shown in panels b–d reflect structural difficulties of the autoencoder in producing certain DSD characteristics (multimodality, noise, sharp peaks). By comparison, the exponential and gamma distribution closures of the 2-moment scheme are able to recover the primary modes in all three distributions relatively well, with the exception of the 90th-percentile example initial condition, where the fixed shape parameters of the distributions lead to a decay in performance.

Figure 4 illustrates the latent-space trajectories of the congestus cloud validation set in 3 dimensions, neglecting the total mass latent variable which is constant under coalescence. Despite the use of distinct time evolution formulations (SINDy, NN-derivative, and AR), the end-to-end trained models encode DSDs to latent spaces with a similar global structure of convergence toward a fixed point, reflective of the nature of coalescence. Indeed, across all three cases, diverse initial conditions which are spread out in the latent space converge toward more similar coalesced final states through a nonlinear evolution, consistent with the nonlinear trajectory found by Will et al. (2023). This consistency suggests that the feed-forward autoencoder identifies an optimal latent geometry for representing coalescence which is largely agnostic to the structure of the time evolution model, despite being trained together.

Finally, we note that some reconstruction error is aleatoric in nature due to the inherent artificial noisiness of the SDM-generated distributions: the stochasticity present in SDM simulations with 100 superdroplets in a $(50 \text{ m})^3$ volume will far exceed that of an equivalent real atmospheric volume containing 10^{13} droplets (100 cm^{-3}). Dziekan and Pawlowska (2017) demonstrated that SDM simulations with order 1,000 super-droplets exhibit 1% variability due to Monte Carlo sampling noise, while those with only 100 superdroplets exhibit 10% variability. An autoencoder that could “correctly” reproduce such noisy DSDs might be considered *overtrained* for realistic DSDs on these 200m-averaged scales.

4.2. Latent Time Evolution

Figure 5 compares the encoded DSD trajectories in latent space with the corresponding predictions from each time evolution formulation (each trained alongside its own autoencoder, and thus its own data encoding). All three models can produce both a near-constant and strongly varying behavior in the latent space, but differ in shape of the trajectory and in susceptibility to noise. Autoencoders associated with the AR and SINDy models present the largest noise and variability within a particular encoded DSD (z_1 for SINDy; z_2 for AR), while the model-predicted latent trajectories are relatively smooth. In general, the NN-derivative formulation generally encodes and predicts smoother trajectories, suggesting that fluctuations in the input DSDs stemming from SDM variability are masked more strongly by the autoencoder associated with the NN-derivative. These findings are corroborated by the somewhat larger reconstruction errors (Figure 3) and intervals (Figure 7) in the NN-derivative case, indicating improved smoothness in time evolution at the sacrifice of some reconstruction precision. SINDy predicts trajectories that are more linear in nature, matching the overall time evolution well for z_2 and z_3 but leading to a larger mismatch in z_1 , where the SINDy model fails to capture the convergence of the variable toward similar values at longer time scales. All three time evolution operators could benefit from an ensemble approach, where multiple networks are trained using different random initializations, and the mean of their predictions is used (Lakshminarayanan et al., 2017).

Mass conservation clearly differentiates the models' learning abilities. SINDy preserves total liquid water mass most effectively, aided by its initialization scheme that starts all polynomial coefficients at zero. By contrast, the NN-derivative and AR models rely on standard neural weight initialization and fail to enforce conservation, with the NN-derivative case showing the largest deviations due to small non-zero mass derivative terms that accumulate under time integration.

These results suggest that flexibility in the time evolution model influences how nonlinearity is distributed between the encoded representation and the time evolution operator. The AR appears to partition the strongest degree of nonlinearity into the time evolution rather than the autoencoder, followed by the NN-derivative case, where the network predicts derivatives rather than discrete time states. SINDy is the most constrained time evolution structure, using only 60 degrees of freedom in its polynomial coefficients (15 coefficients for each of the 4 latent variables) rather than the thousands used by both neural network alternatives. SINDy produces the most nearly constant latent time evolution and benefits from its less-flexible structure in discovering mass conservation. Nevertheless, some degree of time evolution nonlinearity is

Table 1
Hyperparameters Optimized by Optuna by Model Architecture

	AE-AR	AE-NNdzdt	AE-SINDy
Learning rate	✓	✓	✓
Batch size	✓	✓	✓
Layer 1 size	✓	✓	
Layer 2 size	✓	✓	
Layer 3 size	✓	✓	
w_{dx}	✓	✓*	✓*
w_{dz}	✓	✓*	✓*
Latent Dimension			✓
Polynomial Order			✓
w_{dx} Metaweight		✓	✓

Note. If a hyperparameter was excluded from the search space for a network, it is because it is not applicable to that network's architecture. An asterisk (*) next to a checkmark means the hyperparameter was indirectly varied.

essential, with hyperparameter optimization revealing that order-2 polynomials outperform order-1 polynomial SINDy variants (not shown).

4.3. DSD Evolution

Figure 6 demonstrates performance of the complete ROM pipelines including encoding, time evolution, and decoding (in contrast to Figure 3 which demonstrates reconstruction only). Overall, all three models capture the coalescence-driven evolution of the DSD, including broadening, growth of mean droplet size, and emergence or decay of multimodal features. In general, those test members which had higher error in pure reconstruction (Figure 4) also display larger errors in the full pipeline with time evolution, indicating that errors in the autoencoder propagate to the full model performance. Many failure modes are consistent across all three model structures, as reflected in the common patterns of error in Figures 6b–6d. Others show distinct differences. For instance, the 5th percentile—one of the worst performing test cases across model instances—presents a narrow peaked rain mode. The NN-derivative model, and to a lesser extent the AR, predicts an erroneous growth in total liquid water mass. In the 25th percentile case, all models generally predict a slightly broader initial condition, but manage to capture the decay of this mode and the formation of a second peak near 200 μm , with

the NN-derivative standing out for predicting the least mass in this second peak. The 50th percentile case distinguishes models for the rate at which they transfer mass into the second mode, with SINDy slightly underpredicting the size of this second mode, and the AR somewhat overpredicting the size. The NN-derivative model again demonstrates deficiency in mass conservation, predicting a reduction in mean liquid mass despite capturing the DSD shape quite well. The 2-moment scheme struggles to predict certain bimodal initial conditions (5th and 50th percentile) while also suffering from the rapid self-collection issue noted by Seifert and Beheng (2001) that leads to an artificial reduction in overall mass as the rain mode disappears.

These results demonstrate the importance of accuracy in the full model pipeline, from DSD shape and encoding to accurate prediction of latent variable trajectories and mass conservation. Indeed, as we will see in Table 2, the AR

Table 2
Model Performance Comparison Across the Validation Set and Various Test Sets Described in Section 3.1, Taken as an Average and Standard Deviation Across all Set Members and Across Time

Test case “Name” (Purpose)	Metric	AE-SINDy (mean $\pm$ std)	AE-NNdzdt (mean $\pm$ std)	AE-AR (mean $\pm$ std)	SB2 (mean $\pm$ std)
Congestus, 5400s “Validation” (Train/Val.)	KL ($\times 10^{-3}$)	7.3 $\pm$ 11	5.3 $\pm$ 4.9	5.4 $\pm$ 5.3	28 $\pm$ 33
	Wass. ($\times 10^{-3}$)	4.3 $\pm$ 2.3	4.4 $\pm$ 2.2	4.1 $\pm$ 2.2	6.9 $\pm$ 4.0
	Mass MAE	(6.2 $\pm$ 14) $\times 10^{-9}$	(2.9 $\pm$ 3.3) $\times 10^{-2}$	(2.2 $\pm$ 2.7) $\times 10^{-2}$	(1.6 $\pm$ 1.8) $\times 10^{-1}$
Congestus, 4800s “Condensation” (Testing)	KL ($\times 10^{-3}$)	12 $\pm$ 30	11 $\pm$ 19	14 $\pm$ 25	27 $\pm$ 39
	Wass. ($\times 10^{-3}$)	6.2 $\pm$ 4.1	6.9 $\pm$ 4.5	7.6 $\pm$ 4.4	7.9 $\pm$ 5.4
	Mass MAE	(4.5 $\pm$ 22) $\times 10^{-9}$	(3.4 $\pm$ 4.2) $\times 10^{-2}$	(3.4 $\pm$ 6.1) $\times 10^{-2}$	(1.8 $\pm$ 2.4) $\times 10^{-1}$
Congestus, 7200s “Sedimentation” (Testing)	KL ($\times 10^{-3}$)	8.9 $\pm$ 9.3	9.8 $\pm$ 19	10 $\pm$ 11	46 $\pm$ 32
	Wass. ($\times 10^{-3}$)	6.7 $\pm$ 3.8	6.4 $\pm$ 3.5	7.1 $\pm$ 3.9	14 $\pm$ 6.5
	Mass MAE	(1.4 $\pm$ 160) $\times 10^{-9}$	(3.8 $\pm$ 3.8) $\times 10^{-2}$	(2.9 $\pm$ 4.1) $\times 10^{-2}$	(1.4 $\pm$ 1.5) $\times 10^{-1}$
RICO, 20hr “RICO” (Testing, UQ)	KL ($\times 10^{-3}$)	7.5 $\pm$ 5.5	8.8 $\pm$ 7.4	9.8 $\pm$ 8.5	36 $\pm$ 30
	Wass. ($\times 10^{-3}$)	7.1 $\pm$ 4.1	7.5 $\pm$ 4.3	7.9 $\pm$ 4.4	13 $\pm$ 6.0
	Mass MAE	(1.0 $\pm$ 6.3) $\times 10^{-9}$	(3.0 $\pm$ 3.5) $\times 10^{-2}$	(3.0 $\pm$ 5.0) $\times 10^{-2}$	(1.1 $\pm$ 1.6) $\times 10^{-1}$

Note. Best results (lowest values) for each metric and test case are in bold. Note that only the validation split from the 5400s Congestus case are included in the reported results.

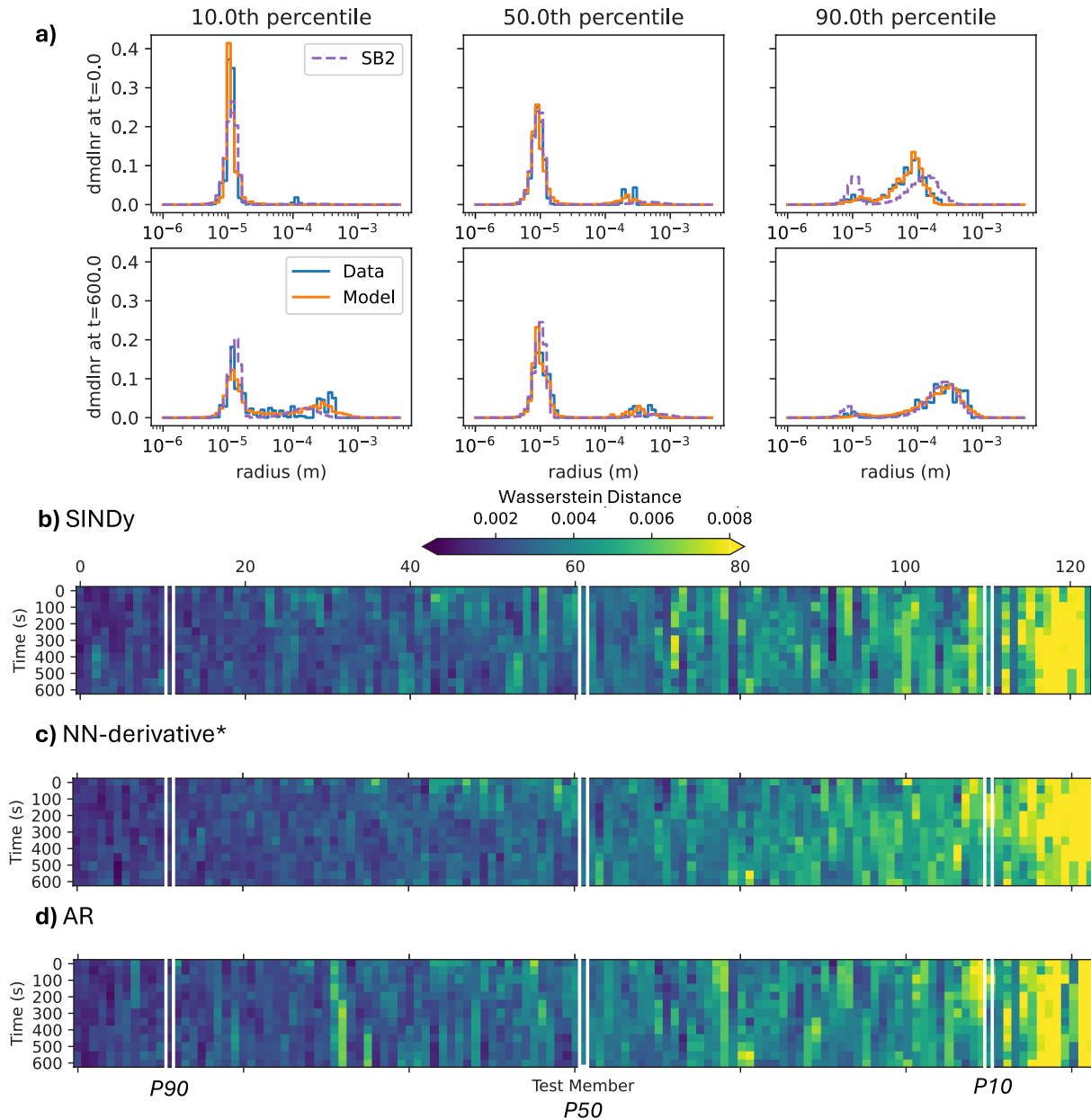


Figure 3. (a) Normalized droplet size distribution reconstructions from the AE-NNdzdt model for three validation members (10th, 50th, and 90th percentile by Wasserstein distance), shown for the initial and final (600s) conditions. 2-moment closures are also included in this panel for illustration. These three test members are highlighted and labeled in (b)–(d) with white boxes. (b)–(d) Reconstruction error between the true and reconstructed DSDs (Wasserstein distance; color) across all validation members over time for autoencoders trained end-to-end with Sparse Identification of Nonlinear Dynamics, NN-derivative, and AR time evolution. * denotes the time evolution model that is depicted in (a). Test members are sorted according to mean error for the NN-derivative, with consistent order across panels (b)–(d).

and NN-derivative cases frequently predict the lowest end-to-end errors when evaluated on normalized DSD metrics, such as the KL divergence or Wasserstein distance, but fall short in accurately capturing mass conservation of the coalescence process, contributing to larger end-to-end errors in the modeling pipeline.

5. Generalizability and Uncertainty Quantification

5.1. Generalizability to Test Sets

In order to assess how well each model formulation generalizes across cloud regimes, Table 2 reports three distinct mean metrics for each formulation on different test sets. Whereas the normalized mass mean absolute

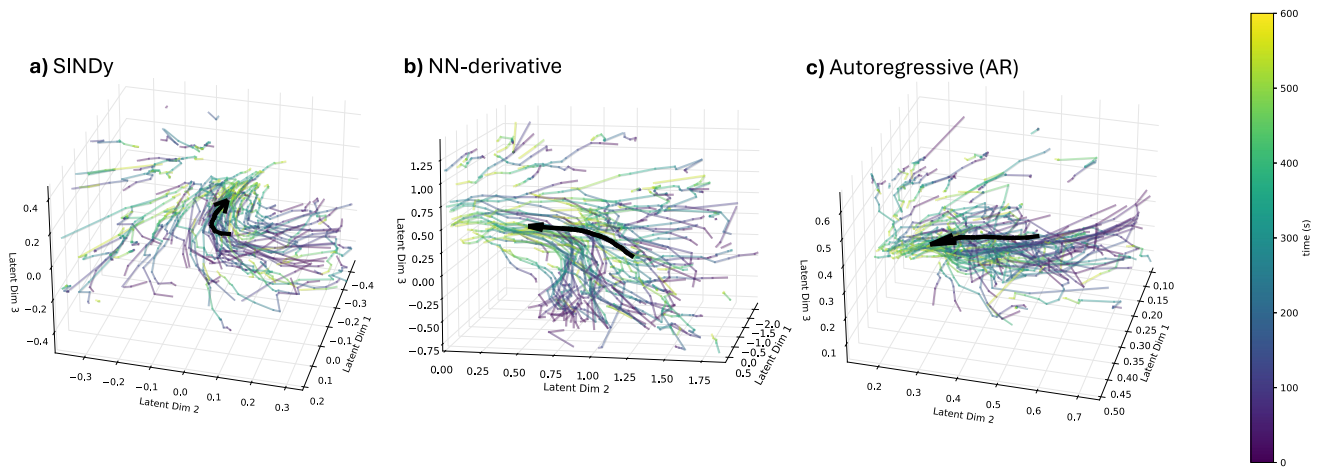


Figure 4. Three-dimensional latent space data using the autoencoder discovered during end-to-end training with (a) Sparse Identification of Nonlinear Dynamics, (b) NN-derivative, (c) autoregressive time evolution. Validation data from the 5400s congestus case are shown, with color indicating time elapsed since the LES was restarted with coalescence-only. Thick black arrow indicates the mean latent space trajectory.

error (MAE) and the KL-divergence between predicted and true DSDs appear in the training loss function directly, the Wasserstein distance, another metric that gauges the difference between two normalized DSDs, is included for further evaluation. Across all test cases and models, the mean error on out-of-sample test data are typically within one-to-two standard deviations of the in-sample validation data set errors, indicating good generalizability of all models to different cloud/rain regimes. Cases dominated by condensation and sedimentation present greater challenges to the data-driven ROMs than does RICO, likely due to the inclusion of more DSDs with very large (sedimentation) or very small (condensation) mean droplet sizes, which were underrepresented in the training and validation sets. This observation suggests that future implementation of this surrogate modeling framework would benefit from incorporating more diverse cloud regimes into the training set.

All three data-driven models achieve comparable errors on normalized DSD predictions across test sets, while the 2-moment parameterization struggles to predict the diverse particle size spectra and evolution, yielding larger spectral errors in KL divergence as well as errors in mass conservation owing to artificially rapid self-collection. SINDy consistently yields the lowest mass errors, as was observed on the validation set (Figure 5). The AR model appears less robust to shifts in data distribution: although it generalizes to other congestus cloud cases comparably well as the NN-derivative model, with a roughly doubling in DSD errors, it experiences a large increase in error when evaluated on the more constrained RICO data set. SINDy, on the other hand, actually displays reduced KL divergence and mass error on the RICO data set. This result suggests that the structurally flexible AR model may be overfit, whereas the second-order polynomials used in SINDy demonstrate greater generalizability.

5.2. Uncertainty Quantification: Conformal Prediction Intervals

Figure 7 demonstrates the uncertainty of the different ROM structures through prediction intervals computed from conformal prediction. Larger prediction intervals indicate a larger degree of uncertainty in model predictions, assessed empirically by aggregating the magnitude of differences between predictions and a calibration data set across various confidence intervals. (Refer to Section 3.5 for a complete description of how intervals are computed, and Figure S4 in Supporting Information S1 for an illustration of the prediction intervals in DSD space.) Reconstruction uncertainty—i.e., the uncertainty associated with the autoencoder alone (shown in Figure 7b)—is comparable in magnitude to the end-to-end model uncertainty. Furthermore, the autoencoders associated with all three models perform comparably in reconstruction. Stronger distinctions emerge from the different representations of latent time evolution (Figures 7c and 7d): SINDy and AR exhibit similar latent-space prediction intervals, while the NN-derivative experiences larger intervals as the evolution operator is integrated forward in time. As in Table 2, the SINDy time evolution operator shows the strongest performance (smallest prediction intervals) in recovery mass conservation (Figure 7d).

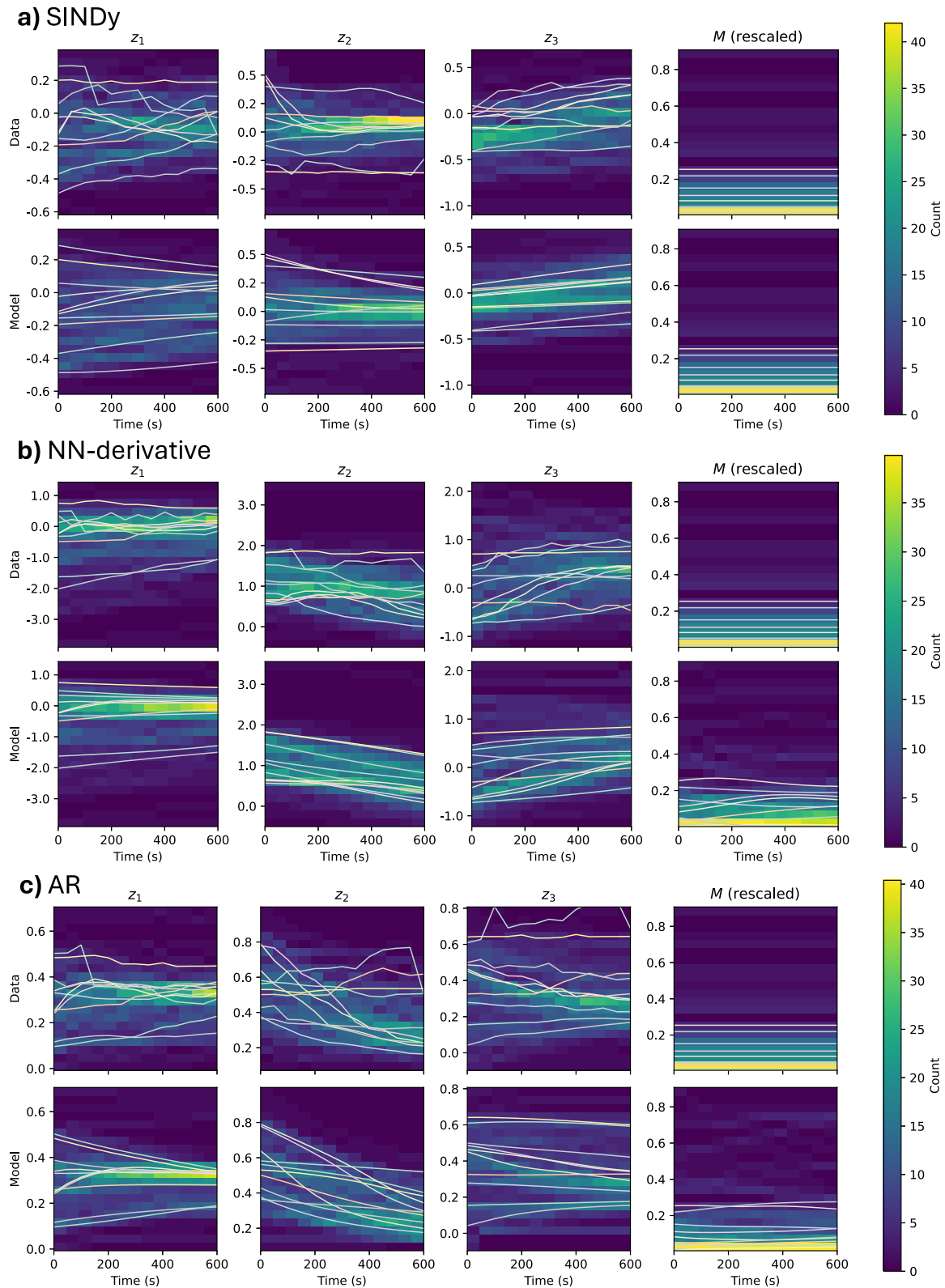


Figure 5.

Prediction intervals on the end-to-end DSD predictions broaden with prediction time regardless of time evolution structure, but with smaller rates of change than the latent variable intervals alone. This behavior indicates that the encoding-decoding step from the autoencoder masks some growth in latent variable uncertainty when converted back to DSD space. Interestingly, the AR uncertainty grows faster with time initially, perhaps due to the single-lagged time-step prediction nature in contrast to a time derivative prediction. SINDy consistently maintains the tightest uncertainty bounds in the end-to-end DSD and mass predictions due to its polynomial initialization that implicitly favors mass conservation, despite its more constrained functional form. The NN-derivative model is in between: with larger latent space uncertainty but smaller variability in mass conservation errors, its end-to-end predictions grow roughly linearly with time but are slightly larger than those for SINDy. These results highlight a key conclusion: greater model flexibility and complexity do not necessarily yield better generalization. The relatively simple SINDy formulation for latent time evolution achieves strong confidence, performance, and generalization, whereas the more expressive neural-network-based structures exhibit reduced generalizability and larger prediction intervals.

6. Discussion and Future Work

This study builds on reduced-order modeling frameworks for microphysics by considering the time evolution of microphysical properties in a data-driven latent space, rather than relying on a typical two-category moment structure. We leverage high-fidelity superdroplet simulations to generate a range of realistic DSDs for model training and evaluation. In addition to the autoencoder and normalization which compress a 64-dimensional binned DSD into 4 latent variables, we compare three candidate approaches for modeling the time evolution of DSDs within this latent space: order-2 polynomials in the SINDy framework, a neural-network-driven time derivative, and a discrete-time autoregressive neural network.

Our results demonstrate that all three time evolution structures are capable of capturing the essential nonlinear evolution of DSDs under collisional coalescence, including broadening, bimodality, and, in some cases, mass conservation. Notably, the SINDy-based approach to time evolution achieved the lowest mass conservation errors and demonstrated superior generalizability across cloud regimes, including out-of-sample test cases such as the RICO simulation, despite having the most constrained model structure. These findings highlight the tradeoff between model flexibility and physical consistency, suggesting that constrained and interpretable time evolution operators such as SINDy, when coupled to a flexible data-driven latent space discovered through an autoencoder, may be preferable to more expressive and complex neural network approaches for representing time evolution. This work extends recent reduced-order modeling approaches introduced by Lamb et al. (2024) and Huang et al. (2025): rather than learning complex ML-parameterized time evolution on simple DSD transforms such as two-category moments, our findings suggest learning *simple* time evolution on *complex* transforms of the underlying microphysical state.

Nevertheless, none of the three reduced-order models are, as presented, ready for operational use. Several additional testing and development steps are underway in order to enable integration of this technique into full atmospheric simulations:

1. *Online testing coupled to traditional microphysics:* Attempts to parameterize physical processes in an offline environment often translate poorly when the same parameterizations are applied online (Lamb et al., 2025), thus there are no guarantees that the offline success and stability of this modeling framework will translate to 3D atmospheric simulations. As a next logical step, the coalescence emulator (for instance, the AE-SINDy model) can be integrated with traditional microphysics such as the P3 scheme (Morrison & Milbrandt, 2015), such that the ROM is called during the coalescence subroutine, and traditional phase-change, sedimentation, and advection processes are used otherwise. This evaluation will require repeatedly mapping between latent variables, DSDs, and two-category moments as in Igel et al. (2022) at every time step, and thus

Figure 5. Comparison of actual versus predicted latent trajectories for each time evolution model (Sparse Identification of Nonlinear Dynamics, NN-derivative, AR), displayed as a heatmap of all validation data. The y-axis is the value of the latent variable displayed for a particular panel, as labeled at the top of each column. Top rows: Ground truth droplet size distribution (DSD) data and total mass encoded into latent space, showing true evolution of the system. Bottom rows: model predictions in latent space, obtained by integrating the learned time evolution model forward in time from the ground truth initial condition. Models predict future latent space states without access to the true DSD data. Light-colored lines display 9 random validation samples for visual reference.

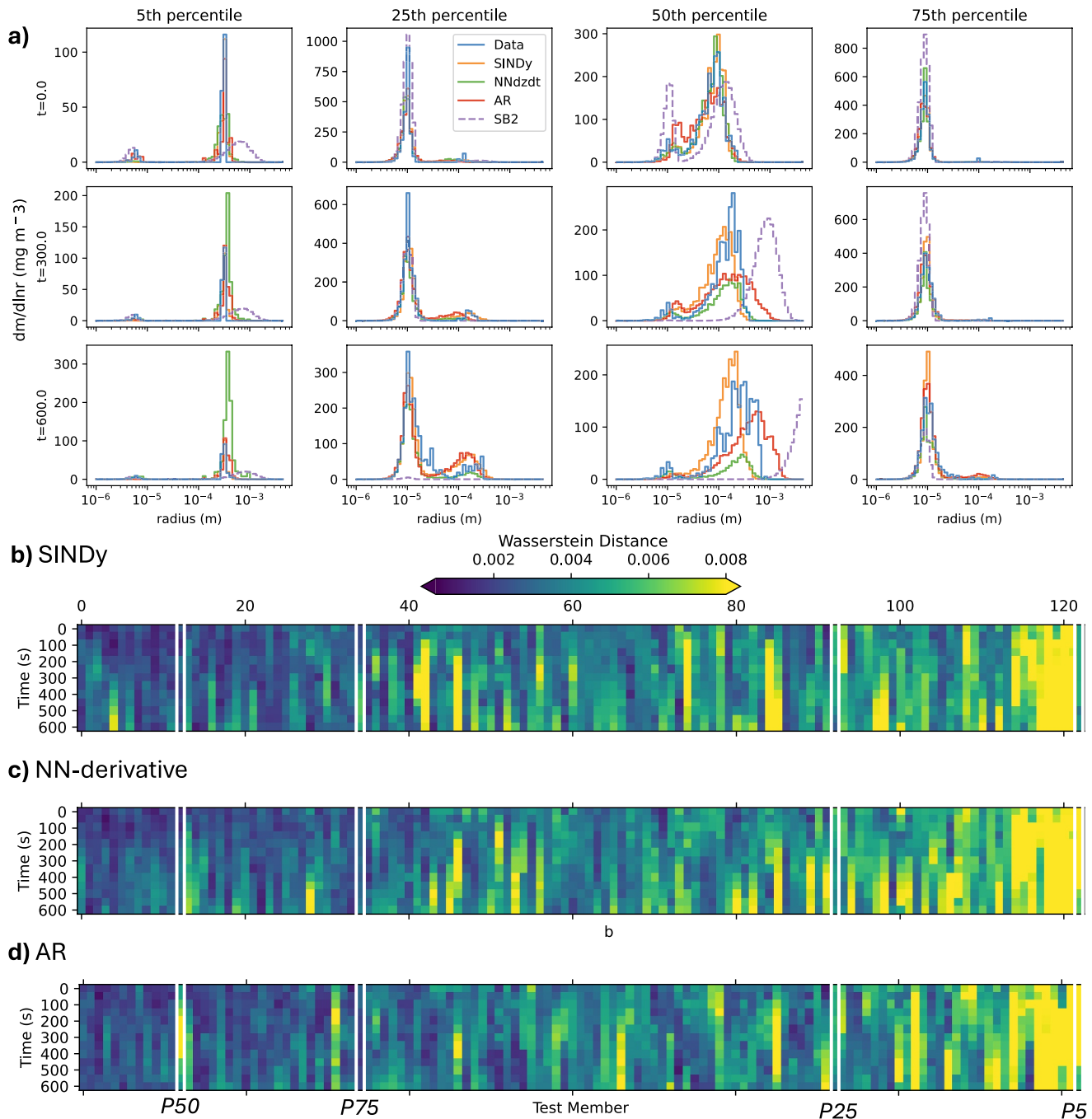


Figure 6. (a) As in Figure 3 for full (dimensioned) DSDs, but using the full model pipelines, including both the autoencoder and time evolution (Sparse Identification of Nonlinear Dynamics, NN-derivative, or AR) components, given only the initial droplet size distribution as input. Test members in (a) are illustrated at three different time steps for all three model structures and are selected based on average performance across all model structures. Panels (b–d) display Wasserstein distance on the normalized DSDs with the full model pipelines. The order of test members in (b–d) is consistent with the order presented in Figure 3.

- is unlikely to be an efficient or operational technique. Rather, it is suggested as a first step to identify and resolve shortcomings, such as numerical instability, in the ROM structure.
2. *Extension to additional liquid-phase processes:* While this study focused on collisional coalescence as a key driver of uncertainty and error in existing parameterizations, the ROM framework can be extended to encompass other processes such as condensational growth and evaporation, particle sedimentation, and advection. As noted by Lamb et al. (2025) and Huang et al. (2025), the use of nonlinear autoencoders prohibits

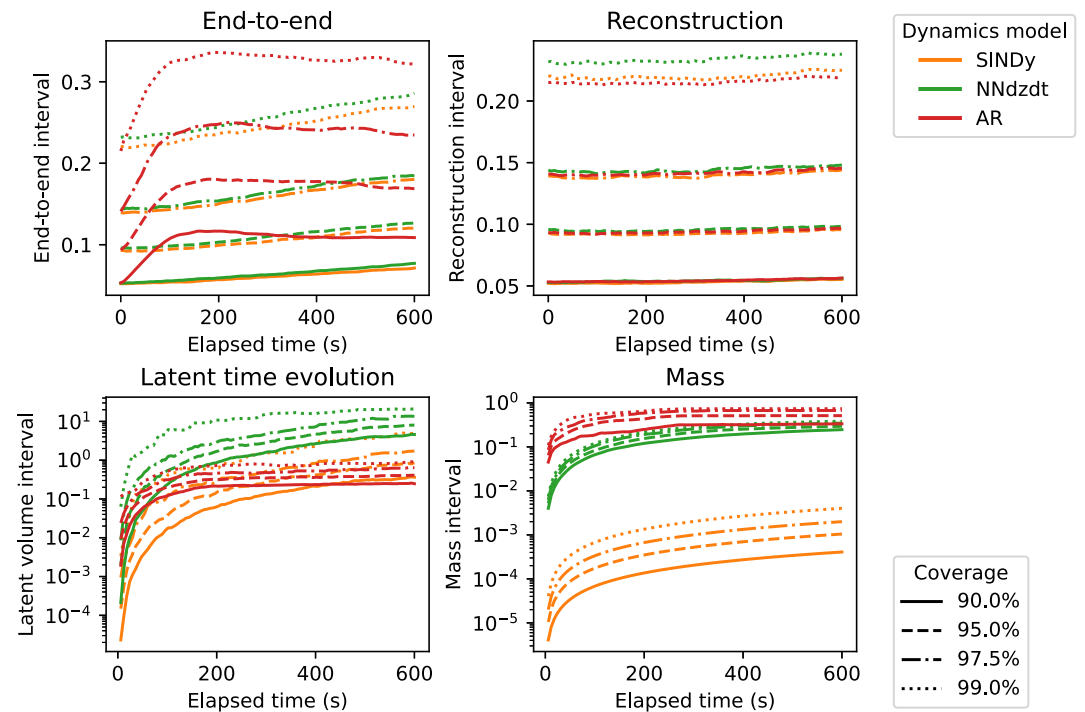


Figure 7. Prediction intervals across time from conformal prediction on the RICO data set for all three time evolution formulations at various coverage levels α , including: (a) normalized droplet size distribution (DSD) predictions from the full reduced-order model pipeline; (b) normalized DSD reconstruction/autoencoder-only; (c) latent-space time evolution only, reported as an elliptical volume; and (d) rescaled mass only.

direct advection or turbulent mixing of latent variables, which are no longer conserved quantities. As a result, a full warm rain ROM must contain an additional advectable latent space encoding which could be learned and optimized similarly from appropriate superdroplet simulations. Likewise, condensation and other particle processes may benefit from different latent space encodings that emphasize the axes of microphysical variation under said process.

3. *Generalization to aerosol and mixed-phase processes:* Most microphysical particles are not, by nature, pure liquid water: aerosols and the many phases of solid water have several axes of variation beyond particle size, and thus will necessitate more complex encodings of a multi-dimensional microphysical state. Unfortunately, high-fidelity training data for these more complex and diverse particles are limited, with SDM implementations of mixed-phase clouds still in development, relying on uncertain parameterizations of many processes such as heterogeneous freezing and neglecting others such as secondary ice production. Ice-phase representations in the SDM still must make parametric assumptions about the core properties of ice particles, such as number of monomers and a spheroid shape (S.-I. Shima et al., 2020), which may lead to bias in a model trained on these data. High-fidelity microphysics simulations and laboratory measurements must continue to advance in order to enable a future unified data-driven microphysics parameterization.

This work sets the stage for data to shape subgrid-scale parameterizations, but with a cautionary reminder that such models can only learn the data that they see. The next frontier in data-driven atmospheric modeling will require combining the richness of high-fidelity simulations with the constraints offered by atmospheric observations, ultimately uniting laboratory, simulation, and satellite perspectives in a common parameterization framework.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All code and the post-processed data used in this study are archived and publicly available (de Jong et al., 2026).

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